

Named Entity Recognition with a MEMM

Ethan Reinhart CS 510 NLP, Spring 2025

May 28, 2025

Abstract

I present a Named Entity Recognition (NER) system using a Maximum Entropy Markov Model (MEMM) under the BIO scheme for PER, ORG, LOC, and MISC. Our streamlined pipeline includes feature extraction, MEMM training with OpenNLP, and evaluation on held-out data. I report token accuracy (97.24%) and entity-level F_1 (84.79%), and discuss a brief ablation study.

1 Introduction

Named entity recognition identifies text spans as people, organizations, or locations. It is essential for tasks such as answering questions and extracting information. I model NER as sequence labeling with nine BIO tags and train a first-order MEMM to capture both rich token features and tag transitions.

2 Background

MEMMs are discriminative sequence models combining logistic regression with a Markov assumption on labels: $P(y_i|x, y_{i-1})$ is computed via MaxEnt, and Viterbi decoding finds the best tag sequence. They allow arbitrary features (unlike HMMs) but normalize locally (unlike CRFs).

3 Implementation

3.1 Feature Extraction

I extract for each token:

- **Lexical:** raw and lowercase form.
- **Orthographic:** capitalization, digits, hyphens, shape, length bin.
- **Morphology:** prefixes/suffixes (1–3 chars).
- **Syntax:** POS, chunk tags.
- **Context:** prev/next word and POS, POS bigrams.
- **Sequence:** previous tag.

3.2 Feature Enhancement

I provide further features for each token by using the following additional resources:

- **brown-c1000-freq1:** the most common words found in the Brown corpus
- **LargestCity:** a list of large cities for proper noun tracking.
- **dist.all.last:** a list of common surnames for proper noun tracking.

3.3 Training and Decoding

I created a decoder to extract the features for the pos-chunk data. This allows for the model to train on richer feature information. First, this was trained and tested using the defined feature extractions. Next, this was trained and tested with the surname and city data. Finally, this was trained and tested with all listed feature enhancements.

4 Results

Table 1 compares test-set performance for the three feature-set iterations you ran.

Table 1: MEMM raw accuracy with different features

Feature Set	Accuracy
Baseline features	97.21%
+ Surname & city	97.21%
+ Brown	97.24%

Table 2: MEMM performance metrics comparison

Feature Set	Precision	Recall	F_1
Baseline features	84.64%	84.59%	84.62%
+ Surname & city	84.65%	84.55%	84.60%
+ Brown	84.80%	84.77%	84.79%

We see that the enrichment of proper noun features provides insignificant performance increase. I hypothesize this is due to the fact that this is already accounted for in the shape feature and word location (first word or not). We see that the addition

of the Brown Corpus data provides a statistically significant testing improvement. I suspect this is largely due to the word frequency count, allowing the model to have better intuition regarding unseen words.

5 Conclusion

I built a MEMM-based NER tagger, achieving 97.24% raw accuracy F_1 . Future work includes CRF extension or character-level dense features to push performance metrics (recall, precision, F_1) beyond 90%.

6 Running the Code

The model can be trained by running `sh train.sh`. The dev testing output can be seen by running `sh dev.sh`. The testing output can be created by running `sh test.sh`. The output of running test will be in maxent directory.